Open-ended module: additional training on data analysis

Examples 1–5 of the syllabus · module marks 5, doubled to 10

HOW THIS PAPER WORKS

The syllabus gives the open-ended module 5 marks. This paper gives **twice** that (10 marks). Each example from the syllabus appears once. You answer one example from each part.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match.

Similar papers used: CSC1MN106 Computer Fundamentals with MS Excel, SPSS (Nov 2024); STA2MN101 Probability Theory I (Apr 2025).

Time 1 hour

Maximum 10 marks

Write complete programs. Each program must read or create its data, process it, and show a plot or print a result. Use the labels and file names given in the question.

Part A · Files and plotting

Answer one of the two. Each carries 5 marks. Total 5.

1. Data file creation (example 1). Write a Python program that generates a CSV file `pendulum.csv` with two columns, `Length` and `Period`, for lengths 0.25 m, 0.50 m, 0.75 m and 1.00 m. Use three period readings per length (values of your own choosing, close to the theory $T = 2\pi\sqrt{L/g}$). Build the data as a pandas DataFrame and save it with `to_csv`, without the index column.

Instructor · Example 1

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2. File read and plot (example 2). Write a Python program that reads `pendulum.csv` (the file from example 1), computes the mean period for each length using `groupby()`, and uses Seaborn to draw a regression line of mean period against length. Label the axes and give the plot a title. Write one sentence on what the plot says about the relation between period and length.

Instructor · Example 2

Part B · Merging and visualisation

Answer one of the two. Each carries 5 marks. Total 5.

3. Merge and group (example 3). Two DataFrames are given. `springs` has columns `spring_id`, `material` and `force`, `extension`. `materials` has columns `material` and `youngs_modulus`. Write code that (a) merges the two tables on `material`, (b) computes the spring constant `k = force / extension` for each row, and (c) finds the mean spring constant for each material using `groupby()`. Explain why the merge step is needed before the group step.

PYQ CSC1MN106 N24 A8 · adapted · Example 3

4. Seaborn visualisation (example 4). Using the table from example 3, draw the linear relationship between the force applied and the extension with `sns.regplot`. Also draw, in the same program, a pandas histogram of the extension column and a pandas scatter plot of force against extension. Explain in two sentences how the regression line shows Hooke's law.

PYQ STA2MN101 A25 B16 · adapted · Example 4

COVERAGE

SYLLABUS ITEM	QUESTION
Example 1: data file creation with a DataFrame	1
Example 2: read the file, mean period, regression line	2
Example 3: merge and group by, spring constant with material	3
Example 4: regplot of force against extension	4
Pandas histogram and scatter plot	4
Seaborn visualisation tools	2, 4

Total marks: 5 (Part A) + 5 (Part B) = 10. Bar charts in pandas are covered in Module 3, question 2.